Amazon UK Product Analysis

Amazon EDA Iron Hack Group – November 4th, 2025

Executive Summary

This report presents a data-driven exploration of Amazon UK's product listings, focusing on product categories, pricing, and customer ratings. The goal is to identify key market trends and business insights that can guide pricing strategies, product focus, and marketing priorities.

Business Context

Amazon UK operates in a highly competitive e-commerce landscape. Understanding which product categories dominate the marketplace, how prices are distributed, and how customers rate their purchases provides valuable input for optimizing listings and increasing profitability.

1. Category Analysis

The frequency analysis shows that certain product categories appear far more often than others, highlighting the dominant market areas. The top five most listed categories include Sports & Outdoors, Home & Kitchen, and Electronics. Sports & Outdoors products lead the listings, indicating strong consumer demand and wide product variety.

2. Pricing Analysis

Average product prices on Amazon UK show a right-skewed distribution, with most items priced lower but a few high-end products significantly increasing the mean. A high standard deviation and wide range confirm strong price dispersion. Outliers were observed beyond £35,000, reflecting luxury or specialized items.

3. Ratings Analysis

Customer ratings display high consistency, with a mode of 5.0 stars and low variance, indicating generally positive customer satisfaction. The slight right skew in the distribution confirms that most ratings cluster toward higher values.

4. Business Implications

These insights suggest that focusing on high-rated and high-demand categories such as Sports & Outdoors may enhance profitability. Monitoring pricing outliers could help identify untapped premium markets. Maintaining high product ratings remains crucial for sustaining consumer trust and conversion rates.

Conclusion & Next Steps

This analysis confirms that Amazon UK’s marketplace is dominated by a few product categories, with diverse pricing and consistently high customer satisfaction. Further analysis could explore correlations between pricing, reviews, and sales volume to uncover deeper business opportunities.